

Sample output — synthetic input, no real person.

Generated from a constructed whole-genome VCF: 200,023 chip-equivalent variants plus 8 rare missense variants in genes the model has cost anchors for, scored offline against AlphaMissense. **No human genome, and no personal health data, was used to produce this document.** The figures are whatever that synthetic file happens to trigger — the point of the sample is the method and the reporting, not the result. This is an illustrative decision-analytic model, not a formal economic evaluation, and not clinical or financial advice.

Health economics

Every economic result from this run, in one place. Source: `synthetic_wgs.vcf`

What is on this page

1. [Pooled cost-effectiveness](#) — the payer view: findings combined once per condition, adherence charged, every correction shown beside the figure it replaced
2. [Your economic sheet](#) — the same genome from your own perspective, health value kept separate from cash
3. [Per-finding detail and cohort projection](#)

1 • Pooled cost-effectiveness

Value of Information — Health Economics VOI

The return-on-health question, in the author's discipline: **what is knowing your genome worth, in expectation?** A decision-analytic model over every actionable finding — discounted net monetary benefit, cost-of-illness averted, and pharmacogenomic averted-adverse-reactions — with a Monte-Carlo sensitivity analysis.

Expected value of your genome
(net of test cost)

\$9,859

95% CI \$4,283 — \$18,224 · P(cost-effective @ \$100k/QALY) = 100%

Marginal value of upgrading chip
→ whole genome

\$3,475

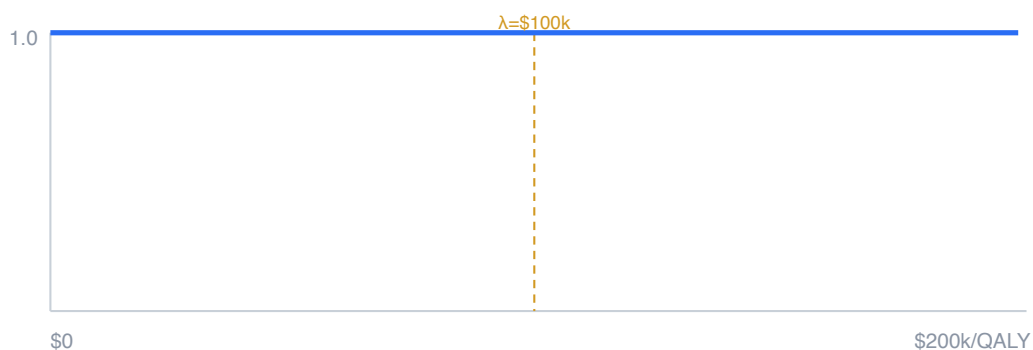
added expected value the WGS-only findings unlock

Downside risk — a bad 1-in-20 outcome

\$4,124

$CVaR_{95}$ (expected shortfall) · VaR_{95}
= \$4,938 — the average and threshold of the worst 5% of simulated outcomes

Finding	Δ QALY	Cost averted	Net monetary benefit	Conf.
APOE ϵ 4 carrier (elevated cardiovascular / Alzheimer's risk)	0.00	\$3,869	\$3,514	moderate
PTPN22 R620W carrier — autoimmune screening awareness	0.00	\$2,579	\$2,509	moderate
predicted-pathogenic 17:7674220 · WGS-only	0.01	\$1,713	\$2,341	higher
predicted-pathogenic 12:21178615 · WGS-only	0.01	\$1,055	\$1,134	moderate
CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)	0.00	\$300	\$490	high
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)	0.00	\$120	\$210	high
SLCO1B1 Intermediate Function (IM-like) (simvastatin)	0.00	\$60	\$90	high
Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency prevention	0.00	\$219	\$-216	low



Cost-effectiveness acceptability curve — P(net benefit > 0) vs willingness-to-pay.

Pooled cost-effectiveness analysis

costs and QALYs reported separately

Worth doing — and it saves money

in plain English

Acting on these findings is modelled to improve health and reduce medical spending by about \$4,223 over the period. That is unusual: most prevention improves health while costing money.

WHAT YOU GET

About 12 extra days of healthy life, on average.

WHAT IT COSTS

Acting on these findings costs about \$1,245 and is modelled to avoid about

WHEN IT PAYS FOR ITSELF

The money spent comes back after about 2.3 years in medical costs you avoid.

\$5,468 of medical
spending over 10 years.

How sure is this?

The model was run 1,500 times, each time with slightly different assumptions drawn from the ranges the research supports. Acting came out worth it in about 100 of every 100 runs. The conclusion is not close to the line. Across those runs the overall value ranged from about \$1,435 to \$21,000.

What to actually do, most valuable first

- **Reduce your risk of a serious inherited condition** — \$6,273 of modelled value
3 separate results in your report point at this. Acting on it is modelled to cut that risk by about 36%.
On the model's assumptions, that is roughly 1 person in 14 avoiding the condition altogether.
- **Reduce your risk of dementia** — \$1,516 of modelled value
One of your results flags this. Acting on it is modelled to cut that risk by about 10%.
On the model's assumptions, that is roughly 1 person in 64 avoiding the condition altogether.

How often it makes a difference

- If the model's assumptions hold, roughly 1 in 14 people with this pattern of results would avoid a serious inherited condition by acting on it.
- If the model's assumptions hold, roughly 1 in 64 people with this pattern of results would avoid dementia by acting on it.
- If the model's assumptions hold, roughly 1 in 333 people with this pattern of results would avoid heart attack or stroke by acting on it.

An average across many people like you, not a promise to you. Most people get nothing from any single preventive action; a few get a great deal, and the average spreads that across everyone.

What would change this answer

Most of what drives this answer is guesswork, not measurement. The biggest single driver is how much acting on a genetic risk actually reduces it — a number nobody has measured directly for most of these findings. Treat the figures as a considered estimate, not a forecast.

These are model estimates for someone with your genetic results, using population-average figures. They are not medical advice, not a prediction about you specifically, and not a substitute for talking to a clinician.

Findings are grouped by the condition they bear on, combined on the risk scale rather than added, and each condition is charged its cost of illness once.

Which number is the answer? This one. The report shows three totals because they answer three questions: this pooled analysis is the *reference case* — what acting on the findings costs and gains. The *value-of-information* figure above it prices something different: what it was worth knowing any of this, which is why it is smaller and why resolving one uncertain parameter can move it. The *personal economic-impact* page itemises the same findings line-by-line for readability; its total runs higher because per-finding figures come from the curated module tables, which are not yet on the provenance registry.

Double-counting correction applied

Adding every finding's risk reduction separately would have claimed **\$14,692** of avoided cost. Several findings speak to the same condition — a polygenic score, a biomarker and a causal estimate for one trait are re-measurements of one liability, not independent chances to prevent three separate events. Combining them on the risk scale and charging each condition's cost once gives **\$5,468**, removing **\$9,225** (62.8%).

Adherence: trial efficacy vs. real-world effectiveness

Every effect size above is measured in people who followed a protocol. Roughly half of people stop long-term preventive medication, and sustained behaviour change fares worse than that. Charged against the benefit, that gap costs **0.015 QALYs** and **\$4,838** of avoided cost — **32.7%** of the health benefit the trials promise. The ongoing intervention cost is discounted by the same factor, because someone who stops taking a statin stops paying for it; the one-off \$300 test cost is not. Because it is spread over fewer realised QALYs, it is the part of this analysis most sensitive to how many people follow through.

WHO (2003), Adherence to Long-Term Therapies: Evidence for Action. Screening uptake and behavioural maintenance are declared assumptions, varied in the sensitivity analysis.

INCREMENTAL COST

-\$4,223

healthcare sector, negative = saves money

INCREMENTAL QALYS

0.032

health gain, not monetised here

ICER

**not defined —
strategy dominates
or is dominated**

cost per QALY gained

NET MONETARY BENEFIT

\$7,410

at \$100,000/QALY

Verdict: dominant (more health, lower cost)

Condition	Findings	Baseline risk	Risk reduction	Adherence	Cost averted	QALYs	NMB
Pathogenic Carrier Screening	3	20.0%	94% → 52% → 36%	65% screening	\$3,705	0.031	\$6,273
Alzheimer Genotype	1	15.0%	30% → 10%	35% lifestyle	\$1,630	0.001	\$1,516
CAD Wellness Prediction	1	20.0%	3% → 2%	50% pharmacological	\$132	0.000	-\$79

Sequencing already found what an array could not

prospective · before testing

Sequencing is expected to turn up a serious, actionable inherited finding that your array missed in roughly 1 person in 56. Averaged over everyone, that is worth about \$491 against an extra \$200 to buy — so on these numbers it is worth it, by about \$291 per person.

► **Flagged in this genome — 2 sequencing-only finding(s), \$3,475 modelled net benefit.**

Sequencing has already found 2 findings in this genome that a genotyping array cannot produce, carrying \$3,475 of modelled net benefit between them. The prospective figures below describe the population this genome was drawn from — roughly 1 person in 56 carries such a finding. For this input that question is already settled: this genome is one of them.

Chance of a serious actionable finding your array missed	1 in 56
Value if you are that person	\$25,189
Expected value, averaged over everyone	\$491
of which pharmacogenomics (mostly already on your array)	\$38
Extra cost over an array	\$200
Net expected value	\$291

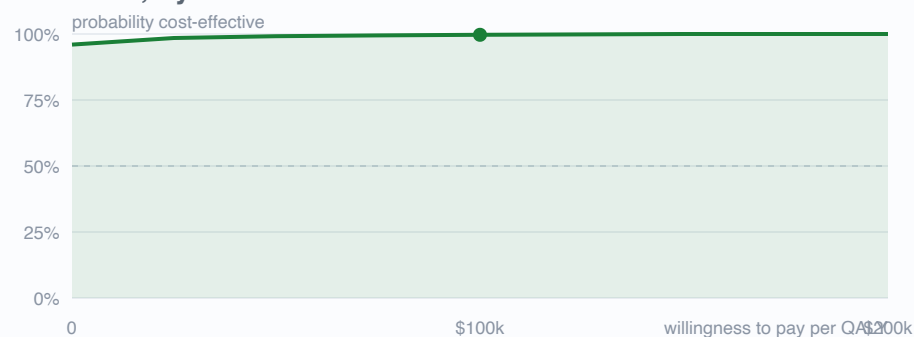
Where the value comes from. Most of this comes from arrays being nearly blind to monogenic disease rather than from sequencing finding exotic things. An array types a fixed list of positions; a normal result for a gene it 'covers' rules out only the specific variants it looks at.

This is an average over a population, and the distribution is extremely skewed: the great majority of people get nothing, and a small number get something that changes their care substantially. An expected value is the right basis for a policy decision and a poor guide to a personal one — if you would want to know, the average does not capture that.

Each iteration draws every parameter with a documented spread from its own distribution and re-runs the model. Net monetary benefit **\$7,759** (95% interval \$1,435 to \$20,594); cost-effective in **100%** of simulations at \$100,000/QALY, cost-saving in **96%**.

Parameters without a published spread are held fixed; the interval below therefore understates true uncertainty rather than overstating it.

Probability cost-effective, by threshold



\$0/QALY	96%
\$25,000/QALY	98%
\$50,000/QALY	99%
\$75,000/QALY	100%
\$100,000/QALY	100%
\$150,000/QALY	100%
\$200,000/QALY	100%

What drives the answer — swing in net benefit across each parameter's range

Parameter	Range	Swing
baseline_event_probability [assumption]	0.05 – 0.45	\$12,229
coi_pathogenic_generic [derived]	30000 – 300000	\$10,004
actionable_rrr [assumption]	0.05 – 0.55	\$5,748
marginal_cost_fraction [derived]	0.4 – 0.8	\$3,645
baseline_event_probability_dementia [assumption]	0.03 – 0.35	\$3,608
adherence_screening [assumption]	0.45 – 0.85	\$3,424
horizon_years_personal [assumption]	5 – 30	\$2,879
correlated_signal_penalty [assumption]	0.25 – 0.8	\$2,586
coi_alzheimer [derived]	100000 – 400000	\$2,445
intervention_cost_standard [derived]	100 – 2000	\$2,395

A parameter high on this list is one the conclusion depends on. Anything tagged [assumption] is a judgement call carrying real weight — worth disagreeing with first.

Judgement, not evidence, is driving this result. 62% of the total swing in net benefit comes from parameters with no published anchor — chiefly baseline_event_probability , actionable_rrr . The conclusion is therefore about as reliable as those judgements are, and a reader who disagrees with them should expect a materially different number. They are stated in full under parameter provenance below.

What is it worth to stop guessing?

Perfect information about every parameter would be worth **\$1** per person (\$878,611 across 1,000,000 people over 10 years). The current recommendation is wrong in **0.2%** of simulations. A figure this small does not mean the model is precise — it means the uncertainty does not straddle the decision. You would act the same way either way, so resolving it changes nothing you would do.

Value of resolving this parameter alone	Per person
baseline_event_probability [assumption]	\$0
coi_pathogenic_generic [derived]	\$0
actionable_rrr [assumption]	\$0
marginal_cost_fraction [derived]	\$0
baseline_event_probability_dementia [assumption]	\$0
adherence_screening [assumption]	\$0

How wrong could each assumption be before the answer changes?

- **baseline_event_probability** [assumption] — Net benefit keeps its sign across the whole plausible range (0.05–0.45 probability), so the recommendation does not turn on this parameter.
- **coi_pathogenic_generic** [derived] — Net benefit keeps its sign across the whole plausible range (30000–300000 \$ per case), so the recommendation does not turn on this parameter.
- **actionable_rrr** [assumption] — Net benefit keeps its sign across the whole plausible range (0.05–0.55 relative risk reduction), so the recommendation does not turn on this parameter.
- **marginal_cost_fraction** [derived] — Net benefit keeps its sign across the whole plausible range (0.4–0.8 fraction of average cost), so the recommendation does not turn on this parameter.

Which strategy, not just whether to act

Strategy	Cost	QALYs	ICER	Status
Genotyping chip	-\$4,423	0.032	—	on frontier
No testing	\$0	0.000	—	dominated

Recommended at this threshold: **Genotyping chip**. Costs are net of averted cost, on the same basis as the incremental-cost figure above. Strategies are ordered by cost. 'Dominated' costs more for no more health; 'extendedly dominated' is beaten by a mix of its neighbours — a check often skipped, which leaves unchoosable options on the frontier.

No whole-genome-only findings are present in this input, so a sequencing arm cannot be estimated from it — only asserted. It is left out rather than invented.

Affordability is reported separately, under the payer budget-impact analysis further down — peak **\$0.5490** per member per month in year 2. Cost-effectiveness and affordability are different questions and the conventions differ, so they are not merged.

Does it work the same for everyone?

Net benefit for Pathogenic ranges from **\$26,532** (Female, age 30) down to **\$4,232** (Male, age 80) — a spread of \$22,300. Illustrative, at 1.0% annual incidence — this shows how the value of prevention changes with age, not a personalised risk estimate. Differences across age come mostly from competing mortality: a prevented event is worth less to someone less likely to survive to experience it. Equity: the Gini coefficient of remaining health moves from 0.216 to 0.217, so the programme **slightly widens** the gap.

The figures above discount the whole horizon at its midpoint, which cannot express the fact that a person has to survive long enough to collect a prevented event. These rerun the top conditions through a three-state Well/Diseased/Dead cohort model against US life-table mortality, with Simpson's 1/3 within-cycle correction. Where the two disagree, the structural figure is the more careful one.

Condition	Δ cost	Δ QALY	Δ life-years	ICER
Pathogenic	-\$2,088	0.217	0.083	—
Alzheimer	-\$1,081	0.049	0.019	—
CAD	\$409	0.009	0.003	\$46,945/QALY

The reference case counts healthcare-sector costs only (\$5,468 averted). The societal perspective adds \$1,815 more, itemised below so that a reader who rejects any one of these valuations can subtract it rather than having to discard the whole figure.

Productivity \$614 QALYs gained × \$55,000/yr × 0.35 working-life share

Unpaid caregiving (dementia) \$1,201 Dementia cases averted × 1,500 h/yr × \$17/h × 3 yr

Sector	Item	Healthcare	Societal	Note
Formal healthcare	Disease treatment cost averted	included	included	Marginal, not average, cost — see marginal_cost_fraction.
Formal healthcare	Test and intervention cost	included	included	Charged once per condition, not once per finding.
Patient	Health-related quality of life	included	included	Reported as QALYs; monetised only at a stated threshold.
Patient	Time and travel for care	not counted	not counted	No defensible per-person estimate available here.
Informal caregiver	Unpaid caregiving time	not counted	included	Dementia only; replacement-wage valuation.
Productivity	Paid labour output	not counted	included	Partial — working-life share stated, not assumed to be 1.
Other	Reproductive decisions	not counted	not counted	Deliberately never monetised; surfaced as a recommendation.
Other	Insurance / discrimination risk	not counted	not counted	Real but not quantified; noted as a limitation.

Internal consistency checks plus one external cross-validation against published statin cost-effectiveness analyses. A model that has never been asked whether it obeys its own constraints is not validated by having a lot of methods in it.

PASS Combined risk reduction respects its cap

largest pooled RRR 0.358 vs cap 0.60

PASS Baseline risks are probabilities

every pooled baseline risk lies in [0, 1]

PASS No condition is charged more than one cost of illness

cost averted per condition never exceeds its full cost of illness

PASS Pooling reduces rather than inflates the total

pooled \$5,468 vs naive \$14,692

PASS Cross-validation: statin primary prevention ICER is plausible

strategy dominant (cost-saving), consistent with published analyses of generic statins in at-risk primary prevention — published analyses of generic statins in primary prevention generally report cost-saving to ~\$50,000/QALY

PASS Most parameters carry a citation

75.4% sourced, 16 declared assumptions

Of the 65 *registered* parameters — method conventions, cost-of-illness anchors, effect sizes and utilities — **8** are read directly from a cited source, **41** are derived from one by a stated arithmetic step, and **16** are judgement calls with no published anchor. The last group is listed in full rather than blended in with the others — they are the parts of this model most worth arguing with.

Whole-model provenance. The percentage above describes the registry only. Counting the **317** per-finding figures in the curated module tables as well — cost, outcome value, prevalence and QALY fields — gives 382 parameters in total, of which **46.9%** carry a citation you can follow directly (PMID or DOI) and **95.8%** carry at least a named literature attribution. Only **4.2%** rest on judgement alone, and those are the declared assumptions listed below.

The remaining gap is **187** figures whose source is named but whose identifier has not been verified, across 52 distinct sources. They are real citations, not invented numbers — they just cannot be resolved automatically yet.

- Kaufman et al. (2014) *Circ Cardiovasc Genet* — 9 figures
- Manchanda et al. (2015) *J Clin Oncol* — 6 figures
- Nherera et al. (2011) *Heart* — FH CEA — 6 figures
- Hicks et al. (2015) *Clin Pharmacol Ther* — CPIC codeine; Bousman (2023) CPIC SSRI — 5 figures
- Innocenti et al. (2009) *J Clin Oncol* — UGT1A1 irinotecan dosing — 5 figures
- Schackman et al. (2008) *Ann Intern Med* — cost-saving — 5 figures
- Kranzler & Soyka (2018) *JAMA* — pharmacogenomics of AUD treatment — 4 figures
- Crist et al. (2018) *Mol Psychiatry* — OPRM1 mu-opioid receptor PGx — 4 figures
- Chen et al. (2012) *Am J Psychiatry* — CHRNA5 × cessation pharmacotherapy — 4 figures
- Hindocha et al. (2019) *Transl Psychiatry* — FAAH rs324420 — 4 figures
- Costa et al. (2013) *Annu Rev Pharmacol Toxicol* — PON1 — 4 figures
- Economopoulos & Sargentanis (2010) *Cancer Epidemiol* — GST × cancer — 4 figures

Declared assumptions

- **chip_pgx_coverage** = 0.85 fraction

Share of the clinically actionable pharmacogenomic variation a consumer array already captures. The main CPIC star alleles sit at common typed positions, so sequencing adds comparatively little here — which matters, because pharmacogenomics is the highest-prevalence category and would otherwise dominate a prospective estimate of what sequencing is worth. Judgement: coverage varies by array and by gene, and CYP2D6 in particular is poorly resolved by both technologies.

- **inequality_aversion** = 11.0 Atkinson parameter

Strength of preference for reducing health inequality in the distributional analysis. Values around this magnitude

appear in the English distributional cost-effectiveness literature, which elicits far stronger aversion than economic intuition suggests; this is recorded as a judgement rather than cited because the figure is carried here from memory of that literature and has not been verified against a specific paper. The result is sensitive to it, so it is reported as an input.

- **subgroup_annual_incidence** = 0.01 probability per year

Annual incidence used for the illustrative age-and-sex subgroup table. The table exists to show how competing mortality changes the value of prevention with age, which it does at any plausible incidence; it is not a personalised risk estimate and is labelled as illustrative in the report.

- **baseline_event_probability** = 0.2 probability

Probability, over the modelled horizon, that a person with an elevated-risk finding experiences the condition it points to, where no condition-specific estimate is available. A generic stand-in for a quantity that genuinely varies by condition, age and sex; it drives every dollar on the benefit side, so it is varied widely in sensitivity analysis rather than defended.

- **baseline_event_probability_dementia** = 0.15 probability

As above, for dementia specifically, set lower than the generic default to reflect that the horizon ends before most of the lifetime risk is realised. Judgement, not an epidemiological estimate.

- **actionable_rrr** = 0.3 relative risk reduction

Risk reduction achievable by acting on a genomic finding where no trial-specific effect is available. Chosen to sit near the measured statin primary-prevention effect (0.27) as a plausible order of magnitude for a well-executed preventive response. This is the single most influential assumption in the model; the range spans doing appreciably less and appreciably better than a statin.

- **intervention_cost_predicted_variant** = 800.0 \$

Cost of following up a computationally predicted pathogenic variant. No standard pathway exists — that is the point of the category — so this is a judgement about what a cautious confirmatory workup would cost.

- **max_combined_rrr** = 0.6 relative risk reduction

Ceiling on the combined risk reduction attributable to acting on all genomic findings for a single condition. No trial delivers elimination of a common complex disease, and without a cap the complement-of-products combination drifts toward 1.0 as findings accumulate. Set at roughly twice the best single-intervention effect in the model (statins, 0.27). Judgement, not a published figure — varied 0.4–0.8 in sensitivity analysis.

- **correlated_signal_penalty** = 0.5 multiplier

Second and subsequent findings routed to the same condition are largely re-measurements of one underlying liability — a polygenic score and a PheWAS biomarker for the same trait are not independent evidence. Each additional finding on a condition is down-weighted by this factor compounding, so the marginal contribution of the fifth CAD signal is small. Judgement; the alternative in use before this was implicit independence, which is certainly wrong in the anti-conservative direction.

- **horizon_years_personal** = 10.0 years

Reporting horizon for the personal economic summary. Chosen for interpretability rather than derived: long enough for prevention to accrue, short enough that the projection is not fantasy. The state-transition model runs to age 100 internally and is truncated to this window for display.

- **adherence_screening** = 0.65 proportion

Uptake of a recommended screening or surveillance programme (colonoscopy, mammography, ferritin monitoring). Higher than chronic drug therapy because the ask is episodic rather than daily, but well short of complete: US population screening uptake sits in the 60-70% band. Anchored on that band rather than on a single study, so it is declared as judgement.

- **adherence_lifestyle** = 0.35 proportion

Sustained behaviour change — diet, exercise, alcohol reduction — maintained over the horizon. The lowest of the three archetypes because maintenance, not initiation, is what the model needs, and maintenance attrition in behavioural trials is severe. Judgement; varied 0.20-0.55.

- **adherence_default** = 0.5 proportion

Fallback for a condition with no archetype assigned. Set equal to the pharmacological figure so an unmapped condition is treated no more optimistically than a mapped one. A condition reaching this parameter is a gap in the mapping, not a finding.

- **coi_coeliac** = 14000.0 \$ per case

Ten-year cost of undiagnosed or poorly-managed coeliac disease: repeat investigation of anaemia and GI symptoms, iron and vitamin repletion, bone-density loss, and the documented gluten-free food premium once diagnosed. Much smaller than the cancer and cardiac anchors in this registry because the disease is managed by diet rather than by procedures or biologics. No single costing study is being followed, so this is declared as judgement.

- **qaly_loss_coeliac** = 0.6 QALYs

Quality-of-life decrement over the horizon from untreated coeliac disease — fatigue, GI symptoms and the consequences of malabsorption. Set well below the cancer and dementia decrements because the condition is controllable once identified. Judgement.

- **penetrance_coeliac_given_dq** = 0.03 proportion

THE FIGURE THAT KEEPS THIS FINDING HONEST. Derived as population prevalence divided by carrier frequency: coeliac disease affects about 1% of people of European ancestry, while roughly 30-40% carry an HLA-DQ2.5/DQ8 susceptibility haplotype. So the haplotype is necessary but nowhere near sufficient, and only a few percent of carriers ever convert. Applying a coeliac cost of illness to every carrier — the error the raw curated figures make elsewhere in this model — would overstate the finding by more than an order of magnitude. Tiered as an assumption, not derived: the reasoning is sound and the two input figures are widely reported, but neither is carried here with a verifiable identifier, and a tier claiming provenance it cannot show is worse than one that admits the gap. Sourcing this against a prevalence meta-analysis would move it to derived and lift the whole registry's sourced share.

References

- Adams PC, Reboussin DM, Barton JC, et al. Hemochromatosis and iron-overload screening in a racially diverse population. *N Engl J Med* PMID:15858186 doi:10.1056/NEJMoa041534 — coi_iron_overload
- Alarid-Escudero F, Krijkamp EM, Enns EA, et al. A Tutorial on Time-Dependent Cohort State-Transition Models in R Using a Cost-Effectiveness Analysis Example. *Med Decis Making* PMID:35924564 doi:10.1177/0272989X221121747 — cycle_correction
- Alzheimer's Association. 2023 Alzheimer's Disease Facts and Figures. *Alzheimers Dement* PMID:36918389 doi:10.1002/alz.13016 — caregiving_hours_annual_dementia, coi_alzheimer
- Birnbaum H, Pike C, Kaufman R, Marynchenko M, Kidolezi Y, Cifaldi M. Societal cost of rheumatoid arthritis patients in the US. *Curr Med Res Opin* PMID:19908947 doi:10.1185/03007990903422307 — coi_autoimmune
- Centers for Medicare & Medicaid Services, Physician Fee Schedule (office/outpatient evaluation and management codes) <https://www.cms.gov/medicare/physician-fee-schedule/search> — intervention_cost_standard
- Cholesterol Treatment Trialists' (CTT) Collaborators. The effects of lowering LDL cholesterol with statin therapy in people at low risk of vascular disease. *Lancet* PMID:22607822 doi:10.1016/S0140-6736(12)60367-5 — statin_rrr_primary
- Diabetes Prevention Program Research Group. Within-trial cost-effectiveness of lifestyle intervention or metformin for the primary prevention of type 2 diabetes. *Diabetes Care* PMID:12832381 doi:10.2337/diacare.26.9.2518 — cost_dpp_program
- Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW. *Methods for the Economic Evaluation of Health Care Programmes*, 4th ed. Oxford University Press (ch. 7, marginal vs average costing) ISBN:9780199665877 — marginal_cost_fraction
- Greenberg PE, Fournier AA, Sisitsky T, et al. The Economic Burden of Adults with Major Depressive Disorder in the United States (2010 and 2018). *Pharmacoeconomics* PMID:33950419 doi:10.1007/s40273-021-01019-4 — coi_depression

- Grzyski JJ, Elhanan G, Morales Rosado JA, et al. Population genetic screening efficiently identifies carriers of autosomal dominant diseases. *Nat Med* PMID:33020650 doi:10.1038/s41591-020-1093-z — wgs_yield_acmg_secondary
- Haque IS, Lizarin GA, Kang HP, Evans EA, Goldberg JD, Wapner RJ. Modeled Fetal Risk of Genetic Diseases Identified by Expanded Carrier Screening. *JAMA* PMID:27532916 doi:10.1001/jama.2016.11139 — wgs_yield_carrier_expanded
- Knowler WC, Barrett-Connor E, Fowler SE, et al. Reduction in the incidence of type 2 diabetes with lifestyle intervention or metformin. *N Engl J Med* PMID:11832527 doi:10.1056/NEJMoa012512 — dpp_rrr
- Kuchenbaecker KB, Hopper JL, Barnes DR, et al. Risks of Breast, Ovarian, and Contralateral Breast Cancer for BRCA1 and BRCA2 Mutation Carriers. *JAMA* PMID:28632866 doi:10.1001/jama.2017.7112 — brca2_penetrance_population
- Litwin MS, Saigal CS (eds). *Urologic Diseases in America*. NIH Publication 12-7865, US Department of Health and Human Services <https://www.niddk.nih.gov/about-niddk/strategic-plans-reports/urologic-diseases-in-america> — coi_urologic
- Mariotto AB, Enewold L, Zhao J, Zeruto CA, Yabroff KR. Medical Care Costs Associated with Cancer Survivorship in the United States. *Cancer Epidemiol Biomarkers Prev* PMID:32229578 doi:10.1158/1055-9965.EPI-19-1534 — coi_breast_ovarian, coi_colorectal
- Miller DT, Lee K, Abul-Husn NS, et al. ACMG SF v3.2 list for reporting of secondary findings in clinical exome and genome sequencing. *Genet Med* PMID:37347242 doi:10.1016/j.gim.2023.100866 — coi_pathogenic_generic, intervention_cost_monogenic
- National Society of Genetic Counselors, patient FAQ on cost of genetic counselling <https://www.nsgc.org/Policy-Research-and-Publications/Genetic-Testing-and-Counseling> — cost_genetic_counseling
- Neumann PJ, Cohen JT, Weinstein MC. Updating cost-effectiveness — the curious resilience of the \$50,000-per-QALY threshold. *N Engl J Med* PMID:25162885 doi:10.1056/NEJMp1405158 — wtp_per_qaly
- Parker ED, Lin J, Mahoney T, et al. Economic Costs of Diabetes in the U.S. in 2022. *Diabetes Care* PMID:37909353 doi:10.2337/dci23-0085 — coi_t2d
- Retail genotyping price plus US self-pay comprehensive laboratory panel price, retail observation <https://www.questhealth.com/> — cost_analysis_bundle
- Sacks JJ, Gonzales KR, Bouchery EE, Tomedi LE, Brewer RD. 2010 National and State Costs of Excessive Alcohol Consumption. *Am J Prev Med* PMID:26477807 doi:10.1016/j.amepre.2015.05.031 — coi_substance_use
- Sanders GD, et al. Recommendations for Conduct, Methodological Practices, and Reporting of Cost-effectiveness Analyses: Second Panel on Cost-Effectiveness in Health and Medicine. *JAMA* PMID:27623463 doi:10.1001/jama.2016.12195 — discount_rate
- Sullivan PW, Ghushchyan V. Preference-Based EQ-5D Index Scores for Chronic Conditions in the United States. *Med Decis Making* PMID:16855125 doi:10.1177/0272989X06290495 — qaly_loss_autoimmune, qaly_loss_cancer, qaly_loss_dementia, qaly_loss_depression, qaly_loss_iron_overload, qaly_loss_mace, qaly_loss_pathogenic_generic, qaly_loss_substance_use, qaly_loss_t2d, qaly_loss_urologic, utility_healthy, utility_post_event
- Tabák AG, Herder C, Rathmann W, Brunner EJ, Kivimäki M. Prediabetes: a high-risk state for diabetes development. *Lancet* PMID:22683128 doi:10.1016/S0140-6736(12)60283-9 — prediabetes_progression_10yr
- Tsao CW, Aday AW, Almarzooq ZI, et al. Heart Disease and Stroke Statistics — 2023 Update. *Circulation* PMID:36695182 doi:10.1161/CIR.0000000000001123 — coi_mace
- US Bureau of Labor Statistics, Occupational Employment and Wage Statistics, home health and personal care aides (31-1120) <https://www.bls.gov/oes/current/oes311120.htm> — caregiver_replacement_wage
- US Bureau of Labor Statistics, Occupational Employment and Wage Statistics, national median annual wage, all occupations https://www.bls.gov/oes/current/oes_nat.htm — productivity_annual
- US direct-to-consumer 30x WGS list prices (Nebula Genomics, Dante Labs), retail observation <https://nebula.org/whole-genome-sequencing-dna-test/> — cost_wgs
- US direct-to-consumer genotyping list prices (23andMe, AncestryDNA), retail observation <https://www.23andme.com/dna-reports-list/> — cost_chip

- US expanded carrier screening self-pay list prices (Invitae, Myriad Foresight), retail observation <https://www.invitae.com/en/providers/test-catalog> — cost_partner_carrier_panel
- US Food and Drug Administration. FDA authorises, with special controls, direct-to-consumer test for three BRCA1/BRCA2 breast cancer mutations (limitations of array-based reporting) <https://www.fda.gov/news-events/press-announcements/fda-authorizes-special-controls-direct-consumer-test-reporting-three-mutations-brca-breast-cancer> — chip_detection_share_monogenic
- US generic drug cash prices (GoodRx), retail observation <https://www.goodrx.com/> — intervention_cost_pgx
- US generic statin cash prices (GoodRx atorvastatin), retail observation <https://www.goodrx.com/atorvastatin> — cost_statin_10yr
- World Health Organization. Adherence to Long-Term Therapies: Evidence for Action. Geneva: WHO <https://iris.who.int/handle/10665/42682> — adherence_pharmacological
- Wright CF, West B, Tuke M, et al. Assessing the Pathogenicity, Penetrance, and Expressivity of Putative Disease-Causing Variants in a Population Setting. Am J Hum Genet PMID:30665703 doi:10.1016/j.ajhg.2018.12.015 — ascertainment_shrinkage

Health economic analysis plan	Cost–utility analysis of acting on genomic findings versus usual care, individual-level inputs.
Study population	One individual's genotype, with population-average parameters — not a cohort study.
Setting and location	United States.
Comparators	Usual care (no genomic testing).
Perspective	Healthcare sector (reference case), societal reported alongside
Time horizon	10 years for the summary; to age 100 in the state-transition model.
Discount rate	3% for costs and effects; 0%/3%/5% sensitivity.
Selection of outcomes	QALYs and healthcare-sector costs, reported separately.
Measurement of effectiveness	Published relative risk reductions; correlated findings pooled on the risk scale, not summed.
Measurement and valuation of preference-based outcomes	EQ-5D-based utility weights from a US catalogue.
Currency, price date, conversion	US dollars; parameter years recorded per parameter in the provenance registry. No inflation adjustment is applied — a stated limitation.
Rationale and description of model	Three-state cohort model (Well/Diseased/Dead) with life-table background mortality and Simpson's 1/3 within-cycle correction.
Analytics and assumptions	Across 382 model parameters, 47% carry a resolvable citation (PMID/DOI) and 96% carry at least a named attribution; 4.2% are declared assumptions, listed individually. 187 figures have a named source whose identifier is not yet verified — a stated limitation.
Characterising heterogeneity	Willingness to pay varied \$50,000–\$150,000/QALY; age and sex enter through the life table.
Characterising uncertainty	Probabilistic sensitivity analysis with parameter distributions from the registry; one-way tornado.
Characterising heterogeneity in distributional effects	Addressed: results are reported by age and sex through life-table mortality, and a distributional analysis applies Atkinson equity weights to report whether the programme narrows or widens health inequality.
Approach to engagement with patients and stakeholders	Not applicable — no stakeholder engagement was conducted.
Effect of uncertainty	Reported as a cost-effectiveness acceptability curve and expected value of perfect information.
Conflicts of interest	None. Non-commercial personal project.

No ROH-driven change to carrier screening

Standard carrier screening is as informative for you as for anyone else; nothing in your homozygosity pattern raises the expected yield of a larger panel.

Decision this informs: Whether to buy an expanded / ancestry-matched carrier panel rather than a minimal one.

F_{ROH} 0.0000 · long-ROH 0.0000 (0 long runs)

A person's own F_{ROH} reflects their parents' relatedness, not their children's, and no published study maps F_{ROH} onto incremental carrier-panel yield. A dollar figure here would be invented, so the output is a recommendation only.

Kirin et al. 2010 PLoS ONE 5:e13996 (long ROH tracks recent pedigree relatedness); ACMG/ACOG expanded carrier-screening guidance (panel composition and ancestry matching)

For your genome specifically

For this genome specifically: the efficient testing choice is Whole-genome sequencing, optimal in 73% of simulations at \$100,000 per healthy year. The modelled value of \$9,859 sits at about the 96th percentile of the population, and — because the data is re-analysed for free as science advances — over the next 20 years it is worth roughly 99% more than a genome frozen at today's knowledge.

Your efficient testing choice (cost-effectiveness frontier)

Option	Cost	QALYs	ICER	Status
No testing	\$0	20.000	—	efficient
Genotyping chip	\$100	20.064	\$1,566/QALY	efficient
Whole-genome sequencing	\$600	20.099	\$14,388/QALY	efficient

Ruled out as inefficient for you: none.

Whole-genome sequencing

efficient choice · optimal in 73% of simulations

96th

percentile of population value

+99%

appreciation over 20y (free re-analysis)

Personal decision-analytic estimate on illustrative parameters — not medical or financial advice.

Extended economic framings

Health capital (Grossman 1972)

Health is a stock that depreciates faster with age — δ rises from **0.0254** to **0.1459/yr** over the horizon. Information doesn't add health directly; it makes health *investment* more efficient, so its value compounds over the years you have left.

Modelled gain: **0.19** discounted health-capital-years · morbidity threshold deferred $\approx$ **1.0 yr**

Optimal timing (real options)

Sequencing is irreversible, prices fall ($\sim 10\%/yr$) and interpretation improves ($\sim 8\%/yr$) — but the data is a permanent asset you can re-analyse free, so waiting only forfeits protection.

Option value of waiting: **\$0** → test now — waiting forfeits more value than the price decline recovers

Risk-adjusted view

Return multiple **32.9×** the test cost · reward-to-variability **2.77** · certainty equivalent **\$8,576** ($\gamma = 2.0$).

Reward-to-variability is Sharpe-style but has no risk-free benchmark, so it measures dispersion, not excess return. The certainty equivalent uses a mean-variance approximation with $\gamma = 2$ (moderate risk aversion).

Expected Value of Perfect Information

The ceiling on what *any* further research could be worth: **\$58**. Current information already captures **99.4%** of the achievable benefit.

EVPI is the maximum ANY further research could be worth — the value of resolving all remaining uncertainty. It is small here, which is itself the finding: the recommended actions stay optimal across essentially the whole uncertainty range, so the decisions are robust and more data would not change them. Low EVPI means 'act', not 'the analysis is weak'.

Risk preferences (Arrow–Pratt)

A risk-averse person values variance reduction above the expected value. Certainty equivalents — $\gamma=1$: \$9,768 · $\gamma=2$: \$9,678 · $\gamma=4$: \$9,497.

Health information is a variance-reducing asset; that is why insurance markets exist (Arrow 1963).

Statistical corrections applied to the genetics, before any economics

Three independent corrections, each shown against the figure it replaced. All three move risk *down*, which is why they matter: an inflated risk estimate inflates every dollar figure derived from it.

Correction	Before	After	Why
Relative risk → absolute risk Falconer liability threshold	RR 2.567×	15.4%	A relative risk with no base rate reads as alarming; the absolute risk is usually far smaller.
European-calibrated → ancestry-adjusted portability, european	11.3%	11.3%	No adjustment: the score is calibrated in this ancestry, so its full predictive variance transfers. This correction bites for non-European ancestries, where much of it does not.
Naive survival → competing-risk cumulative incidence	79.0%	69.8%	Ignoring the chance of dying of something else inflates a lifetime risk, and every dollar figure downstream of it.

Illustrative inputs: a 90th-percentile score against a 6% population prevalence. The chain shows how much each correction moves a risk of that shape, not this person's risk for a named disease.

Which uncertainty actually matters (EVPPI)

willingness to pay: \$0 · event probability: \$0 · treatment effect: \$0

EVPPI ranks parameters by how much resolving *that one alone* would improve the decision; by construction $0 \leq \text{EVPPI} \leq \text{EVPI}$. Every parameter returns ~zero here, which is a substantive result rather than a missing one: the recommended actions do not flip anywhere in the plausible range of any single input, so buying more precision on any of

Why people still don't test (behavioural)

Normative value (3% exponential): **\$6,441** · with present bias ($\beta=0.7$): **\$4,509** · adoption gap **\$1,932**.

Under prospect theory the certain, up-front test cost is felt ~2.25x more than an equivalent uncertain future health gain, and present bias further discounts prevention that pays off decades away. Together these explain why a test with clearly positive expected value still goes unbought — an adoption problem, not a valuation problem.

them would be wasted spend. This is the same robustness the near-zero EVPI reports, decomposed parameter by parameter.

If people live longer, this is worth more

Scenario	Life expectancy	Realised risk	vs today
2025 baseline	79.0	73.2%	+0.0%
2050 projection	85.0	77.0%	+5.3%
Longevity-advance scenario	95.0	81.0%	+10.8%

Longer lifespans raise realised late-onset risk (less competing mortality) AND lengthen the horizon over which prevention pays off. Both increase the value of genomic information, so the default 95-year assumption is conservative.

Markov cohort model (state-transition CEA)

Strategy	Cost	QALYs	Life-years
Standard care	\$29,387	20.004	22.825
Genotype-guided	\$26,726	20.441	23.147
Incremental	\$-2,661	+0.436	+0.322

ICER n/a · NMB \$46,301 · **dominant (cheaper and more effective)**

Annual cycles · half-cycle correction · $p = 1 - e^{-r\Delta t}$ · age-dependent competing mortality · both costs and QALYs discounted · structural validation passed.

Budget impact — the payer's question

Plan of 1,000,000 members, 80,000 eligible. Undiscounted, uptake phased in — **PMPM** is the decision metric.

Year	Uptake	Tested	Offsets	Net impact	PMPM
Year 1	5%	4,000	\$0	\$1,400,000	\$0.1167
Year 2	12%	9,600	\$864,000	\$1,296,000	\$0.1080
Year 3	20%	16,000	\$2,880,000	\$-160,000	\$-0.0133
Year 4	28%	22,400	\$4,032,000	\$-992,000	\$-0.0827
Year 5	35%	28,000	\$5,040,000	\$-1,960,000	\$-0.1633

Peak PMPM **\$0.1167** (year 1) · 5-year cumulative \$-416,000 · becomes cost-saving within the horizon

Welfare: local vs centralised analysis

Conceding centralised platforms a **+15%** capability advantage (gap \$1,524), local is preferred only if the expected privacy cost exceeds it. Expected privacy cost **\$7,673** (55% chance of exposure over the horizon — a genome cannot be revoked) → S_{local} **\$9,859** vs S_{central} \$3,710: **local analysis preferred**.

Break-even annual breach probability: 0.29% — below this, centralised analysis wins; above it, local does.

Local analysis is preferred when the expected privacy cost exceeds the capability gap conceded to centralised platforms. Because a genome cannot be revoked, its exposure hazard compounds over the whole horizon, so even a low annual breach probability accumulates. A second, larger channel is access: disclosure risk deters testing entirely, and a non-tester forfeits 100% of the value — so local analysis raises social surplus mainly by letting more people participate at all.

Illustrative welfare arithmetic with assumed parameters, not an empirical estimate. The capability premium is deliberately set in favour of the centralised alternative so the comparison is not assumed true by construction.

Information economics. Private knowledge of elevated risk without insurer knowledge creates adverse selection (Akerlof 1970; Rothschild–Stiglitz 1976). Where insurers can underwrite on genotype, the fear of discrimination suppresses testing — information that is socially valuable goes unacquired. In the US, GINA (2008) restricts health-insurance and employment use but NOT life, disability, or long-term-care insurance. *Local-only analysis is an economic design choice, not just a privacy one: it lets the individual capture the decision value of the information without disclosing it into a market that would price against them.*

Market price (what these tests cost to buy)

à-la-carte ≈ **\$1,600** vs one consolidated run ≈ **\$300**.

MARKET PRICE (what these cost to buy) — distinct from the health-economic **VALUE** above.

- Perspective: individual lifetime; decision-analytic net monetary benefit.
- Willingness-to-pay λ = \$100,000/QALY (sensitivity \$50,000–\$150,000; Neumann et al., NEJM 2014).
- Discounting: 3% on both costs and QALYs (sensitivity 0/3/5%).
- Test cost modelled: \$300 (wgs).
- Cost-of-illness figures: ADA, AHA, Alzheimer's Association (illustrative).
- PGx averted-ADR values from published CEA (Schackman 2008; Kazi 2014; Deenen 2016; CPIC).
- Uncertainty via seeded Monte-Carlo PSA (Beta on probabilities, Gamma on costs); the WTP threshold λ is swept deterministically for the CEAC.
- Left-tail risk: VaR_{95} (5th-percentile outcome) and CVaR_{95} / expected shortfall (mean of the outcomes at or below VaR_{95}) — standard risk-management measures, applied here to health-economic value.
- Phase-3 predicted variants down-weighted by predictor confidence.
- Health-capital framing: Grossman (1972) — health as a depreciating capital stock; information raises the marginal efficiency of health investment, so its value compounds over remaining life-years.
- Timing: real-options (Dixit & Pindyck 1994) — test now vs defer, given falling sequencing prices and improving interpretation.
- Risk-adjusted views: ROI multiple, reward-to-variability (Sharpe-style, no risk-free benchmark), and a mean-variance certainty equivalent ($\gamma = 2$).
- EVPI (Raiffa & Schlaifer 1961; Claxton 1999): the decision-theoretic ceiling on the value of any further research, with the probability that perfect information would reverse the recommended action.
- Expected utility (Arrow 1963; Pratt 1964): certainty equivalents and risk premia under CRRA risk aversion, $\gamma \in \{1, 2, 4\}$.
- Penetrance is ascertainment-corrected before entering the economic model (family-ascertained literature estimates overstate risk for incidentally identified carriers; Begg 2002, Gabai-Kapara 2014), and GWAS effect sizes are winner's-curse shrunk (Zhong & Prentice 2008).

- Information-economics caveats: adverse selection (Akerlof 1970; Rothschild–Stiglitz 1976) and genetic-discrimination chilling effects (GINA 2008 does not cover life/disability/long-term-care insurance).
- Reporting follows the CHEERS 2022 checklist in spirit; illustrative only.

Illustrative decision-analytic model — NOT a formal economic evaluation, and NOT financial or medical advice. All figures are probabilistic estimates with wide uncertainty; population parameters may not transfer to any individual (and PRS-derived estimates are attenuated for non-European ancestries). Market price and health-economic value are different things and are reported separately.

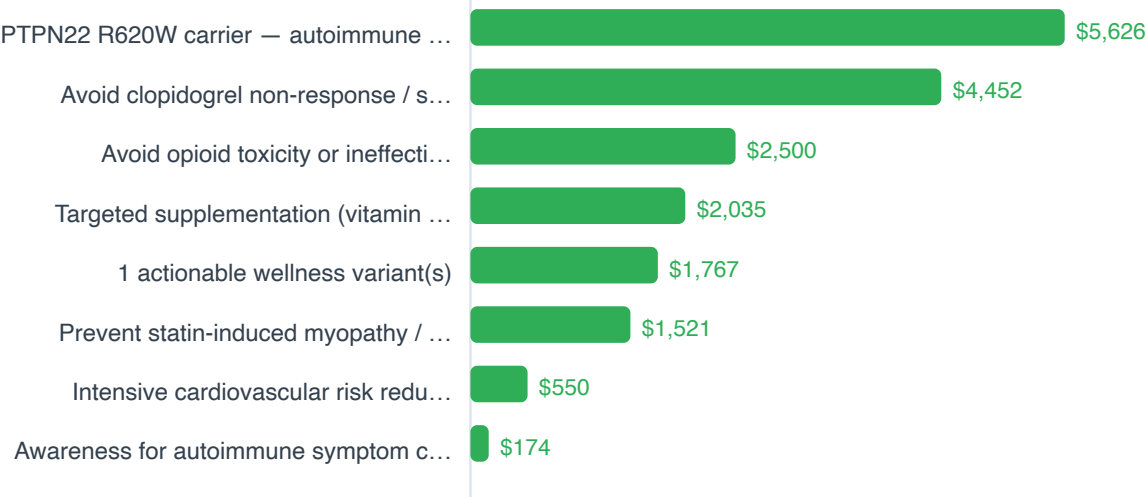
2 · Your economic sheet

Economic-Impact Analysis

A model of the 10-year economic impact of acting on your genomic and blood-work results — expected medical-cost avoidance plus monetised quality-of-life gains, net of intervention cost.

⚠ Illustrative model, not financial or medical advice. Figures use population-average cost-of-illness and risk-reduction estimates applied to your findings; individual outcomes vary widely. Quality-of-life gains are monetised at the standard \$100,000/QALY threshold. Not a guarantee of savings.

Net value by finding



MODELED 10-YR NET VALUE

\$18,625

of acting on your results

plausible range \$9,312 – \$27,938 (±50%)

Medical cost avoided

\$1,221

Intervention cost

\$720

Net *cash* (averted cost – spend)

\$501

Quality-of-life value

\$18,125 (0.22 QALYs)

Value per \$ of analysis (mostly health, not cash)

26.6x

Verdict: cost-saving. Most of the headline figure is *monetised health* (quality-adjusted life-years valued at \$100,000/QALY), not money returned to you. Prevention is typically *cost-effective* — excellent value per healthy year — while still costing more cash than it saves. The two are shown separately so neither is mistaken for the other.

These are real-world figures, not trial figures. Every benefit above has been multiplied by the share of people who actually keep doing the thing — roughly half for daily preventive medication, less for sustained behaviour change, more

for a screening appointment you attend once. Ongoing intervention costs are discounted by the same factor, because someone who stops taking a statin stops paying for it. At full adherence these findings would total **\$36,168**; charging realistic adherence (mean 50%) removes **\$17,543** (48.5%).

WHO (2003), Adherence to Long-Term Therapies. Screening uptake and behavioural maintenance are declared assumptions.

Cost-consequence analysis — costs and consequences, disaggregated

Over 10 years this profile models 0.22 quality-adjusted life-years gained and a net cash position of \$501, from 8 findings. Costs and consequences are listed separately, in their own units, rather than collapsed into a single cost-per-QALY figure — so you can weigh them yourself instead of trusting one ratio.

Type	Measure	Value	Unit	Note
Cost	Testing / analysis cost	\$700	\$ one-off	What you pay to obtain the information.
Cost	Downstream intervention cost	\$720	\$ over 10 yr	Screening, prophylaxis, lifestyle programmes triggered by findings.
Consequence	Medical costs averted (marginal)	\$1,221	\$ over 10 yr	Marginal, not average, cost of the cases avoided.
Consequence	Quality-adjusted life-years gained	0.22	QALYs	Health gain in its own unit — deliberately NOT converted here.
Consequence	Actionable findings identified	8	findings	Count of distinct modelled findings driving the analysis.
Consequence	High-confidence findings	3	findings	Subset resting on stronger evidence.

No single cost-per-QALY figure is given, on purpose. Cost-consequence analysis is NICE's preferred presentation for digital health technologies: it keeps monetary and non-monetary consequences visible side by side instead of hiding them inside an ICER. It is also the appropriate format when the underlying parameters are illustrative. No summary ratio is given by design. Figures are modelled on illustrative parameters — not measurements, not medical advice.

Reported without a dollar figure

deliberately not monetised

These findings are excluded from every total on this page. That is a modelling decision, not a gap.

Reproductive genetics — 1 carrier condition(s) where partner testing would be informative

Decision: Consider partner carrier testing and genetic counselling before conception.

Indicative cost of acting: \$700 · no benefit figure attached

Reproductive outcomes are deliberately not monetised. The cost of testing is shown because it is a real price; no benefit figure is attached because valuing one would price a prospective child and embed one set of reproductive preferences as if it were universal.

Largest avoidable cost: **Targeted supplementation (vitamin D/B12/folate/iron)**

Finding	Cost avoided	QALY (value)	Intervention	Net value
PTPN22 R620W carrier — autoimmune screening awareness Carrier Screening · moderate confidence PTPN22 heterozygous carrier — Awareness for autoimmune symptom clusters.	\$84	0.07 \$5,607	\$65	\$5,626
Avoid clopidogrel non-response / stent thrombosis (MACE) Pharmacogenomic / genomic · high confidence Avoided adverse event × probability of relevant exposure (curated per-condition model).	\$23	0.05 \$4,529	\$100	\$4,452
Avoid opioid toxicity or ineffective analgesia/SSRI dosing Pharmacogenomic / genomic · high confidence Avoided adverse event × probability of relevant exposure (curated per-condition model).	\$13	0.03 \$2,588	\$100	\$2,500
Targeted supplementation (vitamin D/B12/folate/iron) Pharmacogenomic / genomic · low confidence Avoided adverse event × probability of relevant exposure (curated per-condition model).	\$518	0.02 \$1,617	\$100	\$2,035
1 actionable wellness variant(s) Wellness Genetics · low confidence Nutrient metabolism, sleep, and fitness optimization variants with published intervention economics.	\$362	0.02 \$1,510	\$105	\$1,767
Prevent statin-induced myopathy / rhabdomyolysis Pharmacogenomic / genomic · high confidence Avoided adverse event × probability of relevant exposure (curated per-condition model).	\$4	0.02 \$1,617	\$100	\$1,521
Intensive cardiovascular risk reduction (statin) to prevent MI — Intensive statin therapy + lipid monitoring Pharmacogenomic / genomic · moderate confidence Avoided adverse event × probability of relevant exposure (curated per-condition model).	\$165	0.01 \$485	\$100	\$550
Awareness for autoimmune symptom clusters Pharmacogenomic / genomic · moderate confidence Avoided adverse event × probability of relevant exposure (curated per-condition model).	\$52	0 \$172	\$50	\$174

These are estimates based on published pharmacoeconomic literature, not guarantees. Actual ROI varies by clinic, payer, patient adherence, drug mix, and local costs. Intervention costs are one-time for pharmacogenomic tests and annual for lifestyle programs; outcome values are the modeled health-economic value of an averted adverse event. Figures size opportunity for decision support — they are not billing or actuarial values.

3 · Per-finding detail and cohort projection

Health Economics per finding

Per-finding economics, and the same figures scaled to a cohort. Health value and cash are reported separately rather than combined into a single return-on-investment number. Each intervention's cost is weighed against the modeled value of the adverse outcome it averts, with payback period and 3-year NPV (discounted at 3%). The outcome value is credited *once*, at the horizon midpoint: it is the value of preventing one event, and an earlier version accrued it every year, which tripled it.

TOP HIGH-CONFIDENCE INTERVENTIONS

- **CYP2C19 Intermediate Metabolizer (IM) (clopidogrel)** — ROI 1.2:1, value \$300 for \$250 (one-time)
- **CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants)** — ROI 0.6:1, value \$162 for \$250 (one-time)
- **SLCO1B1 Intermediate Function (IM-like) (simvastatin)** — ROI 0.4:1, value \$60 for \$150 (one-time)

What each figure rests on. 1 finding(s) here are valued from the parameter registry — probability of the event, share of it the intervention prevents, and cost of illness, each carrying its own source and provenance tier, and computed the same way as the pooled analysis above so the two agree. 6 are still valued from the curated per-finding tables, which carry a literature attribution but are *not* on the registry: they are part of the 313 unregistered figures, and no anchor exists for their condition yet. 0 are reported with no dollar value at all — a real risk signal with no intervention proven to change the outcome for an asymptomatic carrier. Pricing those would mean inventing an effect size. 4 more are marked *hypothetical / awareness*: the addiction and top-prescribed-drug panels, which were wired at every layer and silently producing nothing until a set of string mismatches was found. They now appear, but they are awareness rather than costed interventions — and the one genuine prescribing decision among them is already priced under pharmacogenomics — so they contribute zero.

FINDING & INTERVENTION	COST	OUTCOME VALUE	QALY GAIN	ROI	PAYBACK	NPV (3Y)	CONFIDENCE
PTPN22 R620W carrier — autoimmune screening awareness Awareness for autoimmune symptom clusters	\$100 ONE-TIME	\$5,000	0.1	50.0:1	1.0 mo	\$4,683	MODERATE
Folate Metabolism (MTHFR) — Nutrient-metabolism variant — deficiency prevention Targeted supplementation (vitamin D/B12/folate/iron)	\$300 ONE-TIME	\$8,000	0.15	26.7:1	1.0 mo	\$7,353	LOW
APOE ε4 carrier (elevated cardiovascular / Alzheimer's risk) Intensive cardiovascular risk reduction (statin) to prevent MI — Intensive statin therapy + lipid monitoring	\$500 ANNUAL	\$2,550	0.045	5.1:1	1.0 mo	\$1,025	MODERATE
CYP2C19 Intermediate Metabolizer (IM) (clopidogrel) Avoid clopidogrel non-response / stent thrombosis (MACE)	\$250 ONE-TIME	\$300	0.35	1.2:1	11.0 mo	\$37	HIGH
CYP2D6 Intermediate Metabolizer (IM) (codeine / antidepressants) Avoid opioid toxicity or ineffective analgesia/SSRI dosing	\$250 ONE-TIME	\$162	0.2	0.6:1	19.0 mo	\$-95	HIGH
SLCO1B1 Intermediate Function (IM-like) (simvastatin) Prevent statin-induced myopathy / rhabdomyolysis	\$150 ONE-TIME	\$60	0.15	0.4:1	32.0 mo	\$-93	HIGH
Below-median longevity composite (improvement headroom) ~43.1 percentile points of headroom to the population median. Reported as a signal, not monetised — the composite aggregates variants already valued individually above.	\$0 ONE-TIME	\$0	0	—	—	\$0	LOW

FINDING & INTERVENTION	COST	OUTCOME VALUE	QALY GAIN	ROI	PAYBACK	NPV (3Y)	CONFIDENCE
CYP2C19 — top-drug actionable (amitriptyline) Avoid clopidogrel non-response / stent thrombosis (MACE)	\$250 ONE-TIME	\$0	0	—	—	—	HIGH
CYP2D6 — top-drug actionable (atomoxetine) Avoid opioid toxicity or ineffective analgesia/SSRI dosing	\$250 ONE-TIME	\$0	0	—	—	—	HIGH
SLCO1B1 — top-drug actionable (atorvastatin) Prevent statin-induced myopathy / rhabdomyolysis	\$150 ONE-TIME	\$0	0	—	—	—	HIGH
CYP2C9 — top-drug actionable (fluvastatin) Prevent major bleeding event from warfarin overdosing	\$300 ONE-TIME	\$0	0	—	—	—	HIGH

Clinic Cohort

100 patients

\$298

Avg cost / patient

\$1,022

Avg benefit / patient

3.4:1

Health value per \$ spent

11.8

QALYs across cohort

Across 100 patients — intervention cost \$29,815, cost averted \$102,206.

Business model, reported separately: \$500/patient/month at 85% gross margin (0.7 months to recover the intervention cost from that margin). This is a commercial assumption, not a health-economic result — it has a different numerator from the figures above, and the earlier version combined the two into one “ROI”.

Cohort Projection

100,000 members with this genomic profile, each finding weighted by its population frequency

This is a projection from one genome, not a plan actuarial estimate: a real plan's members carry different variants. It answers 'what would a cohort like this person look like', which is the question a payer asks about a screening programme, and it is proportional to the clinic block above by design.

100,000

Members affected

\$29,815,000

Total intervention cost

\$102,206,000

Modeled savings

3.4:1

Health value per \$ spent

**withheld —
dominant**

Cost per QALY

\$72,391,000

Net cash

263,000

Intervention-events

263,000 intervention-events across 100,000 members: each finding is counted once per member who carries it, and one member can carry several. The prevalences sum past 1.0, so the event count exceeds the headcount — it is a count of actions, not of people.

Cohort corrections applied

This block used to report the raw curated per-finding figures: every finding counted separately, no pooling of findings that act on one condition, no real-world adherence, no discounting, and average rather than marginal cost of illness. It claimed **\$319,110,000** of savings and **25925.0** QALYs. Applying the same corrections the individual sheet already applies — marginal cost fraction 0.6, midpoint discount 0.8626, mean adherence 0.515, and correlated-target pooling — gives **\$102,206,000**, removing **\$216,904,000** (68.0%).

The old “cost per QALY” of \$2,654 was gross intervention spend divided by QALYs — no cost offsets and no comparator, so not an ICER despite the label. It sat beside a claim of multi-billion savings, which cannot both be true.

Estimates only. These are estimates based on published pharmacoeconomic literature, not guarantees. Actual ROI varies by clinic, payer, patient adherence, drug mix, and local costs. Intervention costs are one-time for pharmacogenomic tests and annual for lifestyle programs; outcome values are the modeled health-economic value of an averted adverse event. Figures size opportunity for decision support — they are not billing or actuarial values.

GENOMELENS-BUILD: 1bc662e+dirty 2026-08-24T15:50:41-04:00